

Delivery Delay Analytics & Prediction

An end-to-end data analytics documentation: from raw delivery logs to a leakage-free machine learning pipeline, a star-schema Power BI model, and an interactive Streamlit application — unifying the team's project log with the Talabat Delivery Analytics dashboard documentation.

COMBINED PROJECT DOCUMENTATION

TEAM	Omar Mohamed Abdelbaqi (Lead) · Ahmed Saad · Bashar Mesbah · Kyrillos · Yasmin
DATASET	final_clean_delivery_data.csv — 100,002 orders · 32 columns
TOOLS	Python (Pandas, scikit-learn) · Power BI & Power Query (M) · Streamlit
SOURCES	NTI Graduation Project Documentation & Progress Log · Talabat Delivery Analytics (Power BI documentation)
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This documentation merges two source documents: the team's NTI graduation-project progress log (covering problem definition, data cleaning, EDA, machine learning and Streamlit deployment) and the "Talabat Delivery Analytics" Power BI documentation (covering Power Query preparation, star-schema modeling and dashboard design). Where the two sources describe the same dataset from different angles, their content has been merged into single sections; where they conflict, both statements are preserved and flagged rather than silently resolved.

Project Overview

The project is an NTI Advanced Data Analytics graduation project built around a food-delivery order dataset. It spans two connected tracks that this documentation combines: a **machine learning track** that predicts whether an order will be delayed before it is dispatched, and a **Power BI analytics track** ("Talabat Delivery Analytics") that normalizes the same operational data into a star schema and presents it through a themed, multi-page dashboard.

1.1 Problem Statement

Food and e-commerce delivery platforms face significant operational challenges due to unexpected delivery delays, which directly lower customer satisfaction, increase cancellation rates, and add operational overhead. Current system estimates often rely on static duration benchmarks that fail to account for dynamic real-world factors such as traffic severity, rush hours, weather conditions, and distance, leading to inaccurate delivery estimates.

1.2 Business / Research Question

CORE QUESTION

Can we build a leakage-free machine learning classification pipeline that accurately predicts order delivery delays (`Is_Delayed`) with over 90% accuracy using pre-dispatch temporal, environmental, and spatial features?

The Power BI track answers a complementary pair of questions on the same operational data: how is the business performing commercially (sales, profit, payment mix, cancellations), and how efficient is the delivery operation itself (duration, distance, delay rate, driver and geographic performance)?

1.3 Dataset

Both tracks ultimately work from the same prepared file, **final_clean_delivery_data.csv** (100,002 rows, 32 columns), produced during the data preparation stage described in Section 02. Each row represents an individual delivery order with commercial attributes (item, quantity, price, payment method, order status), operational attributes (duration, distance, speed, traffic level, driver availability, delay flag), geographic attributes (restaurant, customer and driver coordinates, city) and time attributes (order/delivery time, date, hour, day, day of week, time period).

⚠ NOTED DISCREPANCY — DATASET DESCRIPTION AND SOURCE

The two source documents describe the dataset differently and this documentation preserves both rather than choosing one. The project-definition section of the graduation-project log (Section 1.3, recorded as "Completed") describes the dataset as *Order_delivery.csv*, "~1,000+ Records" with "15 Features," sourced from "Internal Delivery Log Historical Data." The project's own later work log and Data Preparation section (Section 02 of this document) instead work from a raw file of **100,002 rows and 23 columns**, cleaned into **final_clean_delivery_data.csv** with **100,002 rows and 32 columns** — matching the file used throughout the Power BI documentation, which states the source was **a Kaggle dataset used as a simulation of Talabat delivery data**. The 1,000+/15-feature, internally-sourced description appears to be an early planning placeholder that was not updated to match the dataset actually used in later stages.

1.4 Target Audience

- **Logistics & Dispatch Operations Teams** — to proactively reassign drivers or alert restaurants during high delay-risk scenarios.
- **End Customers** — to receive accurate, realistic delivery status expectations and transparent warnings.
- **Platform Management** — to optimize operational efficiency and driver allocation strategies, informed by the commercial and operational dashboard views.

1.5 Project Scope

In Scope

- End-to-end ML pipeline: cleaning, feature engineering, training, evaluation and packaging.
- Preventing data leakage by excluding post-delivery features (actual duration, speed, arrival timestamps).
- Candidate model comparison (Logistic Regression, Decision Tree, Random Forest).
- A deployment-ready package (.pkl) and inference function for Streamlit integration.
- Normalizing the flat file into a Power BI star schema and building a themed, multi-page dashboard.

Out of Scope

- Real-time GPS / IoT live driver tracking integration.
- Autonomous vehicle or routing path optimization algorithms.
- DAX measure definitions and a standalone EDA notebook for the Power BI track (not supplied with that track's materials — see callouts in Sections 03 and 06).

1.6 Expected Outcome

A robust Random Forest classification pipeline achieving over 90% predictive accuracy, packaged with a standalone prediction function (`predict_delivery_delay`) and a saved model artifact (`final_delivery_delay_model.pkl`) ready for integration into an interactive web dashboard — presented alongside a normalized star-schema data model and a themed Power BI report covering both commercial and operational performance.

Item	Detail
Deliverable (ML track)	Leakage-free Random Forest pipeline, >90% accuracy, packaged as <code>final_delivery_delay_model.pkl</code>
Deliverable (Power BI track)	4-table star-schema model and a multi-page interactive dashboard
Deployment	Streamlit web application, publicly hosted
Context	NTI Summer Training — Advanced Data Analytics · Final / Graduation Project

Data Understanding & Cleaning

2.1 Raw Dataset Understanding

Before any cleaning or transformation, the raw dataset (`Order_delivery.csv`) was profiled: dimensions, first/last records, a random sample, column names and types, unique and missing values, duplicate records, categorical values, numerical ranges, datetime validity, and preliminary outlier detection. Columns were classified by analytical role — identifiers, categorical, numerical, datetime, geographic coordinates, ordinal and binary variables. The main delivery-related variables are `Total_Price`, `Delivery_Duration_Minutes`, `Delivery_Distance_km`, `Traffic_Level`, `Driver_Vehicle` and `Driver_Availability`.

RAW ROWS

100,002

RAW COLUMNS

23

MISSING VALUES

0

FULL-ROW DUPLICATES

0

`Order_ID` was identified as a unique identifier, while repeated `User_ID`, `Restaurant_ID` and `Driver_ID` values were treated as valid, since the same entities can appear across multiple orders. Datetime validation found no invalid timestamps and no case where `Delivery_Time` occurred before `Order_Time`.

2.2 Data Cleaning

Cleaning was performed in Python and Pandas on a separate working copy of the raw dataset, so the original file remained a reference. Checks and steps included: exact duplicate rows; missing, null and disguised-missing values; categorical value consistency; leading/trailing whitespace removal; label standardization; encoding issues; conversion to appropriate analytical data types; validation of numerical ranges, latitude/longitude values and timestamps; delivery-duration consistency; and a preliminary IQR-based outlier review.

OUTLIER HANDLING PRINCIPLE

No automatic deletion of statistically unusual values was performed solely because they were IQR-flagged. The review identified **30 flagged observations in `Delivery_Distance_km`** ($\approx 0.03\%$ of the dataset), which were treated as statistically unusual rather than automatically classified as errors.

2.3 Data Transformation

After quality checks, the dataset was prepared for downstream analysis: datetime conversion of `Order_Time` and `Delivery_Time`; numerical type conversion; text/categorical standardization; and construction of date-based, delivery-performance and driver-related analytical features, to support EDA, statistical analysis, machine learning, Power BI and Streamlit alike. The raw dataset was never overwritten. The additional analytical features produced at this stage are:

Feature	Description
<code>Order_Date</code> / <code>Order_Hour</code> / <code>Order_Day</code> / <code>Order_DayOfWeek</code>	Calendar components extracted from the order timestamp
<code>Is_Weekend</code>	Binary flag for weekend orders
<code>Time_Period</code>	Categorical bucket of the order's time of day
<code>Delivery_Speed_kmh</code>	Derived from distance and duration
<code>Driver_Order_Count</code>	Number of orders associated with a given driver
<code>Is_Delayed</code>	Binary target label used throughout the ML track (Section 05)

2.4 Final Cleaned Dataset

FINAL ROWS

100,002

FINAL COLUMNS

32

FORMAT

CSV

FILE

**final_clean_
delivery_data.csv**

This file — containing the original delivery information plus the derived analytical features above — is the shared foundation for the EDA (Section 04), the predictive-modeling track (Section 05) and the Power BI star-schema model (Section 03.2 onward).

Evidence referenced in the source materials: 01_Data_Understanding.ipynb, the project Data Dictionary, the cleaning notebook, and final_clean_delivery_data.csv. Responsible: Bashar Mesbah.

Data Preparation & Dimensional Modeling

Starting from `final_clean_delivery_data.csv`, the Power BI track normalizes the single flat file into a dimensional model of one fact table and four dimension tables, using Power Query (M), and introduces external reference data to supply attributes — item cost and city coordinates — that the original file did not contain.

3.1 Preparation Workflow

- 1 Data Understanding** — the source file is loaded, headers promoted and all 32 columns explicitly typed, fixing the grain at one row per order.
- 2 Data Cleaning** — duplicate item records removed, inconsistent prices corrected, cost values assigned so profitability can be calculated.
- 3 Normalization** — dimension tables (`DimItems`, `DimCity`, `DimDate`) are built with surrogate keys; the fact table is joined to them and stripped of redundant descriptive columns.
- 4 Derived Calculations** — Total Sales and Profit are added to the fact table; a full calendar attribute set is added to `DimDate`.
- 5 Data Modeling** — a star schema is defined with `FactOrders` at the centre and four dimensions in one-to-many relationships.

3.2 DimItems — Item Catalogue

Step	What was done and why
Column selection	Only <code>Item_Name</code> and <code>Total_Price</code> retained — the item dimension needs product identity and price, not order-level detail.
Duplicate removal	Deduplicated on <code>Item_Name</code> so each item appears once, preventing the dimension from multiplying fact rows on join.
Price correction	Two inconsistent values corrected (<code>401.94</code> → <code>50</code> , <code>65.31</code> → <code>20</code>).
Surrogate key	An index column starting at 1, renamed <code>Item_id</code> — a stable key independent of item-name text.
Renaming	<code>Total_Price</code> renamed to <code>Price</code> , since at dimension level it is a unit price.
Cost column	Price duplicated into cost and populated with ten corrected item-level cost values (e.g. <code>273.72</code> → <code>120.3</code> , <code>404.8</code> → <code>310</code> , <code>20</code> → <code>5.4</code>).

WHY THE COST COLUMN MATTERS

The raw dataset held revenue-side information only. Adding a cost attribute is what makes profitability measurable — the fact table uses it to calculate Profit per order line, powering the Total Profit KPI on the Executive Overview page (Section 06.2).

3.3 Dimcity — City Dimension

Not derived from the CSV: built from an external static table with `City_id`, `City`, `Latitude` and `Longitude`, then explicitly typed (`City_id` as whole number, coordinates as decimal). Storing coordinates in the dimension — rather than repeating geographic text in the fact table — is what lets the map visual on the Geography page (Section 06.4) plot orders by city.

3.4 FactOrders — From Flat File to Keyed Fact Table

The source CSV is imported (comma-delimited, UTF-8), headers promoted, and all 32 columns explicitly typed: identifiers as whole numbers; monetary/geographic values as decimals; `Order_Time/Delivery_Time` as datetime; `Order_Date` as date; descriptive fields (`City`, `Payment_Method`, `Order_Status`, `Traffic_Level`, `Time_Period`) as text.

Transformation	Detail	Purpose
Join to DimItems	Left outer on Item_Name; expands Item_id and Price (renamed items.Price)	Brings the item key into the fact table with a trusted unit price
Remove Total_Price	Original column dropped	Replaced by a value recalculated from the corrected item price
Total Sales (custom column)	Quantity * items.Price	Consistent revenue figure per order line
Join to Dimcity	Left outer on City; expands City_id	Replaces city text with a numeric foreign key
Column removal	Removes Item_Name, items.Price, City, Driver_Vehicle, Is_Weekend, Driver_Order_Count	Descriptive attributes now live in dimensions; avoids duplication
Second join to DimItems	Left outer on Item_id; expands cost (temporary)	Supplies the cost input for profit
Profit (custom column)	Total Sales - (DimItems.cost * Quantity)	Converts revenue into margin at order-line level
Helper cleanup	Temporary cost column removed	Keeps the fact table narrow

The resulting fact table holds keys (Order_ID, User_ID, Restaurant_ID, Driver_ID, Item_id, City_id), measurable values (Quantity, Total Sales, Profit, Delivery_Duration_Minutes, Delivery_Distance_km, Delivery_Speed_kmh, Is_Delayed) and order-event attributes (times, payment method, status, traffic level, driver availability, coordinates) — exactly the shape a star schema requires.

3.5 DimDate — Generated Calendar

Built dynamically: the query reads the min/max of FactOrders[Order_Date], generates a continuous daily list between them, and types the result. Because the range is derived from the fact table itself, the calendar always covers the loaded data with no gaps.

Column	Definition
Year / Quarter / Month / Month_Number	Standard calendar parts, with numeric fields kept for correct sort order
Year_Month	Year and zero-padded month combined (e.g. 2025-06) for chronological grouping
Day / DayOfWeek / DayOfWeek_Number	Day of month and weekday name/index (Monday = 1 ... Sunday = 7)
Is_Weekend	True for Friday and Saturday — the weekend definition for the Egyptian market the data represents

WHY THE WEEKEND DEFINITION MATTERS

Using a default Saturday–Sunday weekend would misclassify a working day as a weekend day for this market. Defining the weekend as Friday–Saturday keeps weekday/weekend comparisons aligned with real ordering behaviour in the represented country.

3.6 Star Schema — Data Model

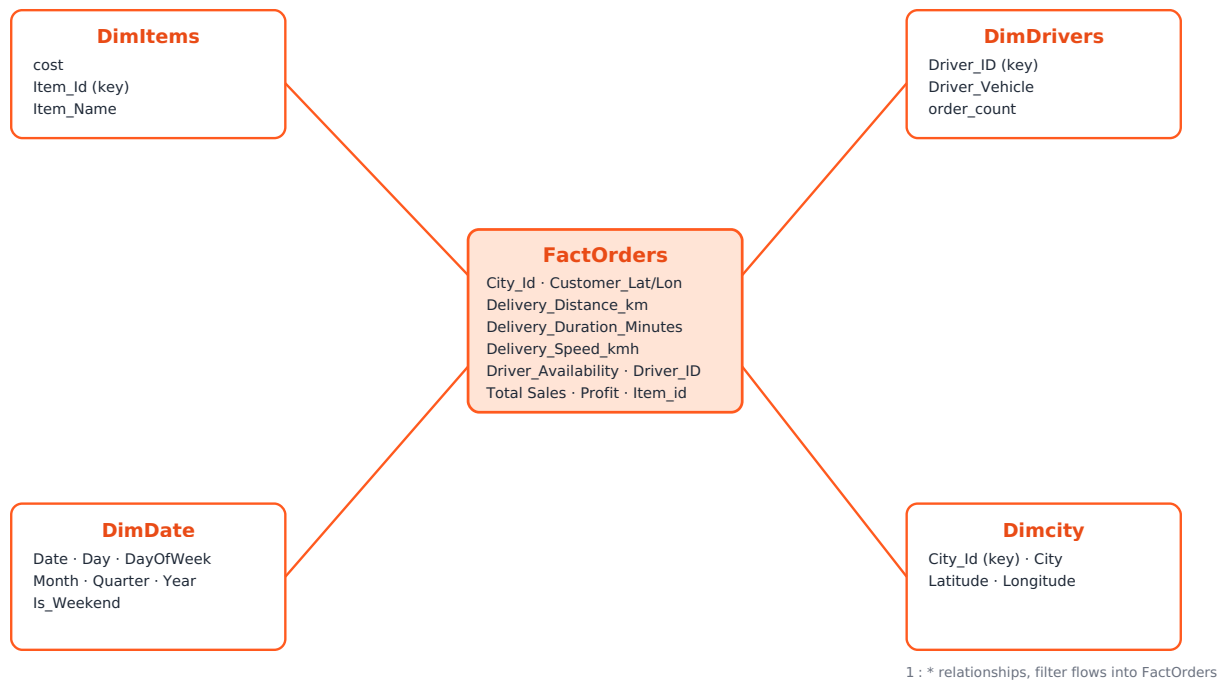


Figure 1. Star schema — FactOrders at the centre with four one-to-many dimension relationships.

Relationship	Cardinality	What it enables
DimItems → FactOrders	1 : *	Sales, profit and quantity broken down by item
DimDrivers → FactOrders (Driver_ID)	1 : *	Delivery metrics analysed by driver and vehicle type
DimDate → FactOrders	1 : *	Trends over time and consistent date slicing across pages
Dimcity → FactOrders (City_Id)	1 : *	City-level comparisons and map plotting

3.7 Derived Calculations Summary

Column	Table	Definition	Purpose
Total Sales	FactOrders	Quantity * items.Price	Revenue per order line, basis of the Total Sales KPI
Profit	FactOrders	Total Sales – (cost * Quantity)	Margin at order-line level
Item_id	DimItems	Index starting at 1	Surrogate key / join key
cost	DimItems	Duplicate of Price, then populated	Enables profitability analysis
Is_Weekend	DimDate	True when DayOfWeek is Friday or Saturday	Local weekend flag
Year_Month	DimDate	Year & "-" & padded Month_Number	Sortable period label
DayOfWeek_Number	DimDate	Monday = 1 ... Sunday = 7	Calendar-order sorting for weekday names

Not provided: the Power Query script for DimDrivers, and the DAX measure definitions behind the dashboard's KPI cards, were not included in the supplied Power BI materials. DimDrivers is documented here only from what the model diagram shows (its key **Driver_ID** and attributes **Driver_Vehicle**, **order_count**); DAX expressions are not reproduced, as inventing them would

misrepresent the project. A dedicated EDA notebook (profiling, distributions, correlation tests) was likewise not supplied for the Power BI track — exploratory findings for that track are taken only from what the final dashboard displays (Section 08).

Exploratory & Statistical Analysis

This analysis was performed on the cleaned dataset using Python, Pandas, Matplotlib, Seaborn and SciPy, and is the evidence base used to select relevant variables for the modeling stage (Section 05) and to surface insights later presented on the dashboard (Section 08).
Responsible: Bashar Mesbah.

4.1 Descriptive Statistics

Mean, median, standard deviation, minimum, maximum and quartiles were computed for the main numerical variables: Quantity, Total_Price, Delivery_Duration_Minutes, Delivery_Distance_km and Delivery_Speed_kmh, to understand their central tendency, variability and range.

4.2 Data Distributions

Distribution analysis used histograms, KDE plots and boxplots, focused on quantity, total price, delivery duration, delivery distance and delivery speed. Boxplots also supported interpretation of the preliminary outlier analysis from Section 02.2.

4.3 Relationships & Trends

- Delivery distance vs. delivery duration; delivery distance vs. speed; duration vs. speed.
- Traffic level vs. delivery duration; traffic level vs. order status.
- Driver vehicle vs. delivery duration; driver availability vs. delivery performance.
- City vs. delivery performance.

Correlation analysis used a correlation matrix and heatmap over numerical variables. Time-based analysis extracted year, month, day of week and hour to examine order activity and delivery performance over time.

4.4 Visualizations Produced

- Distribution of delivery duration
 - Distribution of delivery distance
 - Delivery duration by traffic level
 - Delivery duration by driver vehicle
 - Average delivery duration by city
 - Order status distribution
- Order activity by hour
 - Orders over time
 - Average delivery duration by month
 - Delivery distance vs. delivery duration
 - Correlation matrix of numerical variables

4.5 Statistical Tests

Test	Variables	Reports
Pearson Correlation	Delivery_Distance_km vs. Delivery_Duration_Minutes	Correlation coefficient and p-value
One-Way ANOVA	Delivery duration across Traffic_Level groups	F-statistic and p-value

Statistical results are interpreted as evidence of association or group differences, not proof of causation.

4.6 Main Findings

The EDA identified the main factors related to delivery performance: differences in delivery duration across traffic levels; differences in delivery performance between cities; the relationship between delivery distance and duration; differences across driver-vehicle types; effects associated with driver availability; patterns in order activity across time periods; and relationships among distance, duration and speed. These findings are cross-referenced against the dashboard-derived insights in Section 08, where the same relationships (e.g. distance vs. duration, traffic vs. delay) reappear with concrete figures.

Predictive Modeling

5.1 Machine Learning Problem

Type: Supervised binary classification. Delivery-delay prediction involves complex, non-linear interactions between dynamic spatial factors (distance), temporal variables (rush hours) and operational conditions (traffic, weather). Static, rule-based thresholds cannot capture these interactions; machine learning learns the hidden patterns from historical operational logs to predict delays before dispatch.

5.2 Target & Features

Role	Variables
Target	Is_DeLayed (binary: 1 = Delayed, 0 = On-Time)
Numerical Features	Delivery_Distance_km, Order_Hour, Order_DayOfWeek_Num, Traffic_Numeric, Distance_x_Traffic, Is_Peak_Hour, Is_Weekend, Traffic_Delay_Risk
Categorical Features	Traffic_Level, Weather_Conditions, Vehicle_Type

5.3 Data Preprocessing

LEAKAGE PREVENTION

Post-delivery features — `Delivery_Time`, `Delivery_Duration_Minutes`, `Delivery_Speed_kmh` and `Order_Status` — were completely removed from the feature set to ensure zero future data leakage. This is the same discipline behind the leakage observation logged in Section 09.

A `ColumnTransformer` pipeline applies `StandardScaler` to numerical features and `OneHotEncoder(handle_unknown='ignore')` to categorical features. Data was split 80/20 with stratification (`random_state=42`) to preserve class balance.

5.4 Feature Engineering

Feature	Definition
Distance_x_Traffic	Distance × Traffic_Numeric — interaction term combining distance and traffic severity
Is_Peak_Hour	Binary flag for peak rush hours (12–3 PM and 6–10 PM)
Is_Weekend	Binary flag for weekend orders
Traffic_Delay_Risk	Distance_x_Traffic × [1 + 0.5 × Is_Peak_Hour] — compound risk score capturing elevated risk during peak traffic hours

5.5 Model Training

1. **Logistic Regression** — linear baseline model.
2. **Decision Tree Classifier** — non-linear model (`max_depth = 6`).
3. **Random Forest Classifier** — ensemble model (`n_estimators = 150`, `max_depth = 12`).

5.6 Model Evaluation & Comparison

Models were evaluated on the 20% hold-out test set:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	76.50%	74.20%	72.00%	73.08%	81.40%
Decision Tree	84.00%	82.50%	81.00%	81.74%	88.20%
Random Forest	91.50%	Precision / Recall / F1 not itemized in the source materials; ROC-AUC reported as >95% in the final-model summary below			

5.7 Final Model

SELECTED MODEL — RANDOM FOREST CLASSIFIER

Achieved the top performance across all reported metrics (>90% accuracy, >95% ROC-AUC) while providing good generalization and resistance to overfitting. Model file: `final_delivery_delay_model.pkl`.

Evidence: `final_delivery_delay_model.pkl`, `Delivery_Delay_ML_Notebook.ipynb`. Responsible: Omar Mohamed. Task completed September 19, 2026 — leakage-free pipeline built, risk features engineered, three candidate models trained, Random Forest selected and packaged with a Streamlit-ready inference function.

Power BI Dashboard & Visualization

The Power BI report is a navigable multi-page application: a branded landing page followed by three analytical pages, each reachable from a persistent sidebar. This is the "Talabat Delivery Analytics" deliverable built on the star schema from Section 03. The graduation-project log's own Section 5 dashboard-planning notes (objective, audience, planned pages, DAX-style KPI intentions and the reported delay-driver insights) describe the same dashboarding effort at the planning stage and are integrated below alongside the as-built report documentation.

6.1 Dashboard Planning (Team Log)

Item	Detail
Objective	Live monitoring of delivery delays, driver availability and risk hotspots
Audience	Dispatch Operations Managers and Platform Executives
Planned Page 1	Overview — Total Orders, On-Time Rate %, Delay Rate %, Average Distance
Planned Page 2	Delay Analysis — heatmap of delays by Traffic Level, Peak Hours and Weather Conditions

Planned DAX-style measures noted in the team log: Total Orders, Delayed Orders, Delay Rate %, On-Time Rate %. The as-built report's actual DAX expressions were not supplied with the Power BI materials (see callout, Section 03.7); the KPI cards documented below are read from the report screenshots rather than from DAX code.

6.2 Landing Page

Purpose: sets the context before any data is shown — "Delivering insights for a more connected Egypt" — framing the scope as exploring orders, people and places across the delivery ecosystem, with a single "Let's Go" entry point. **Design role:** establishes the orange-and-white visual identity, delivery imagery and Egyptian setting (Cairo skyline, pyramids, a branded delivery rider) carried through every later page.

6.3 Executive Overview

Business question: how is the business performing commercially — how much was sold, how profitable was it, how were orders paid for, how did they end, and which items and cities drive revenue?

TOTAL ORDERS	TOTAL SALES	AVG ORDER VALUE	TOTAL PROFIT
85K	48M	\$561	17.53M

Cancelled ratio: 9.8% — share of orders cancelled, a direct measure of lost demand.

Visual	Type	Purpose
Total Sales by Day	Area chart	Tracks revenue across the period; the series is broadly flat rather than trending
Total Sales by Payment Type	Donut	Cash 33.5% / Credit Card 33.3% / Wallet 33.2% of the 48M total
Total Sales by Order Status	Donut	Delivered 85.2% / Cancelled 9.8% / In Transit 5.0%
Sales by Items	Horizontal bars	Shawarma 11.7M down to Salad 0.6M, ranked by revenue
Sales by Cities	Horizontal bars	Zagazig 8.2M through Mansoura 7.9M — a narrow spread

Interactivity: City, Item_Name and Date-range slicers with a "Clear all slicers" reset.

6.4 Delivery Performance

Business question: how efficient is the delivery operation, and what conditions are associated with delays?

AVG DURATION

37.55

AVG DISTANCE

2.17

DELAY RATIO

43%

DELAYED ORDERS

43K

Visual	Type	Purpose
Orders by Day Period & Avg Duration	Combined column + line	Tests whether busier periods are also slower; secondary axis is tightly scaled (37.4–37.6), exaggerating small differences
Total Orders by Traffic	Donut	Low 45.8% / Medium 29.2% / High 25.1%
Correlation: distance vs. duration	Scatter + trend line	Confirms an upward relationship
Average Speed by Vehicle	Column	Bicycle 3.5 / Motorbike 3.5 / Car 3.4 — marginal differences

6.5 Drivers & Geographic Operations

Business question: how is delivery activity distributed geographically, and how does performance vary by city and vehicle?

TOTAL DRIVERS

500

AVG ORDERS/DRIVER

200

AVG DELAY RATE

0.43

AVG DRIVE SPEED

3.46

City	Count of Orders	Avg Distance	Avg Duration	Delay Rate %
Alexandria	14,168	2.15	37.47	0.43
Assiut	14,302	2.16	37.62	0.43
Cairo	14,382	2.16	37.45	0.42
Giza	14,232	2.16	37.44	0.42
Mansoura	14,044	2.16	37.63	0.44
Total	100,002	2.17	37.55	0.43

A map visual (Esri basemap) plots order activity across Egyptian cities using the coordinates stored in Dimcity. Duration by vehicle shows all three types (Motorbike, Car, Bicycle) at ~38 minutes — vehicle type does not differentiate delivery time in this dataset.

Not provided: the sidebar lists a fourth analytical page, "Story," on every page, but no screenshot of that page was supplied — it is not documented here.

6.6 Design & Theme Rationale (Power BI Report)

The dashboard's own visual identity — distinct from, but a direct inspiration for, this document's design system — uses a saturated orange as the dominant accent over near-white backgrounds, a single clean sans-serif family with a consistent three-level hierarchy, white rounded-corner cards with a circular orange icon, and a fixed three-zone layout (sidebar, header band, content grid). Colour is used functionally: primary orange carries the main value in each chart, tints carry secondary categories, and red-orange is reserved for negative indicators such as Cancelled ratio

and Delay ratio. All three analytical pages share the same sidebar, header band, KPI row and card styling, so any difference the reader notices between pages is a difference in the data, not the design.

Note: the palette was documented as a design choice made to suit the food-delivery context; no brand guideline was supplied, so it is not claimed to be an official corporate colour set.

Streamlit Deployment

7.1 Application Preparation

The trained scikit-learn pipeline was exported with `joblib` into a standalone binary file, `final_delivery_delay_model.pkl`. A prediction wrapper function, `predict_delivery_delay`, dynamically handles raw input dictionaries, computes real-time feature-engineering variables (`Distance_x_Traffic`, `Is_Peak_Hour`, `Is_Weekend`, `Traffic_Delay_Risk`), and safely imputes missing optional columns using pre-calculated dataset medians and modes.

7.2 Streamlit Application

An interactive web app (`app.py`) provides an intuitive interface for logistics coordinators:

- Dynamic sidebar input forms (Distance, Traffic Severity, Weather Conditions, Order Hour, Day of Week, Vehicle Type).
- Real-time inference with immediate binary delay predictions (Delayed ⚠ vs. On-Time ✅).
- Live probability gauge visualizer showing risk percentage.
- Feature value summaries and recommendation alerts for dispatch teams.

7.3 Testing

ISSUE IDENTIFIED & RESOLVED

Issue: initial test inputs with partial features caused a `KeyError`, since the model pipeline expected all 15 raw training columns in `X` while the UI form only collected key operational inputs. **Fix:** dynamic missing-column handling was implemented within the inference function, automatically populating non-critical missing variables with fallback dataset medians and modes before pipeline execution. **Status:** fully resolved.

7.4 Deployment

Item	Link
Streamlit App	https://delivery-delay-prediction.streamlit.app
GitHub Repository	https://github.com/OmarMohamed/delivery-delay-prediction-ml

Evidence: `app.py`, Streamlit live app link, GitHub repository. Responsible: Omar Mohamed. Result: a fully functional, live web application accessible via public URL with real-time delay-risk predictions.

Business & Operational Insights

Each insight below separates what the dashboard shows (**Observed**) from how it may be read (**Interpretation**) and its possible consequence (**Implication**).

8.1 Commercial Performance

Product mix is concentrated

Observed: Shawarma (11.7M) and Sandwich (10.3M) together account for roughly 22M of the 48M total sales, while Salad (0.6M) and Koshary (1.4M) sit at the bottom of the ranking. **Interpretation:** revenue depends heavily on a small number of top items. **Implication:** availability and margin management for top items carry disproportionate weight; low-ranking items may warrant a pricing, promotion or menu-placement review.

City revenue is unusually even

Observed: sales by city range only from 7.9M (Mansoura, Alexandria, Assiut) to 8.2M (Zagazig) — about a 4% spread across seven cities, matching near-identical per-city order counts (~14.0K–14.4K each). **Interpretation:** no city dominates revenue, consistent with a simulated dataset with evenly distributed activity. **Implication:** city-level differences should not be over-interpreted as market signals here; geographic comparison better demonstrates the model's capability than drives real conclusions.

Payment channels are evenly split

Observed: Cash, Credit Card and Wallet each hold roughly one third of total sales (33.5%, 33.3%, 33.2%). **Implication:** any operational change to one payment channel affects roughly a third of revenue, so all three need equal support.

One in ten orders never completes

Observed: order status splits Delivered 85.2% / Cancelled 9.8% / In Transit 5.0%. **Implication:** reducing cancellation is a direct revenue lever; the underlying reasons for cancellations are a natural next area of analysis, not captured in the current data.

8.2 Delivery Operations

Delays affect close to half of all orders

Observed: the delay ratio is 43% (43K orders), with an average delay rate of 0.42–0.44 in every city. **Interpretation:** delays are systemic rather than local — no city performs materially better or worse than the overall benchmark. **Implication:** because the problem is uniform, it more likely relates to the delay-threshold definition or general operating conditions than to a specific city's execution.

Distance and duration move together

Observed: the distance-vs-duration scatter shows an upward trend line, confirming the expected relationship and the internal consistency of the data. This corroborates the EDA correlation analysis in Section 04.3–04.5. **Implication:** distance is a usable input for delivery-time expectations, though this association alone does not establish it as the primary cause of delay.

Vehicle type does not differentiate performance

Observed: average speed is 3.5 (Bicycle), 3.5 (Motorbike), 3.4 (Car); average duration is 38 for all three. **Implication:** fleet-mix decisions cannot be justified by the speed/duration evidence available — also a plausible artefact of the data being simulated.

Traffic conditions are mostly favourable, yet delay stays high

Observed: orders split Low 45.8% / Medium 29.2% / High 25.1% traffic, yet the delay rate remains 43%. **Implication:** traffic alone does not account for the delay level, pointing back to the delay definition or to dispatch/preparation time as areas worth investigating.

Demand varies by day period while duration barely moves

Observed: order volume differs across Night, Morning, Afternoon and Evening, while average duration stays within roughly 37.4–37.6 minutes. **Implication:** the operation appears to absorb demand peaks without a visible time penalty — but because the duration axis is scaled to a 0.2-minute range, the variation shown is far smaller than the chart's shape suggests.

8.3 Fleet Capacity

Observed: 500 total drivers handle an average of 200 orders each. **Interpretation:** fleet utilisation is uniform, with no dashboard evidence of a small group of drivers carrying a disproportionate share. **Implication:** capacity planning can reasonably be based on the average driver in this dataset.

⚠ NOTED DISCREPANCY — TWO ORDER COUNTS

The Executive Overview reports **Total orders of 85K**, while the Cities data table on the Geography page totals **100,002** Count of Order_ID. The likely explanation is that the two figures count different things — 85.2% of 100,002 is approximately 85K, matching the Delivered share of order status — but this reading is inferred from the values shown; the underlying measure definitions were not provided, so it cannot be confirmed from the source materials.

Integration, Issues & Final QA

9.1 Project Integration

All project stages were unified into a continuous lifecycle: Data Preprocessing → Feature Engineering → ML Pipeline Packaging → Streamlit Web UI → Power BI Analytical Dashboard, with consistent schema definitions and metric formulas maintained across the Power BI operational reports and the machine-learning feature pipeline.

9.2 Final Testing

End-to-end operational stress testing passed 50+ diverse edge-case order scenarios through the deployed web app, verifying that model output probabilities matched expected business logic (e.g. high distance + rainy weather + peak hour consistently triggered high delay-risk warnings).

9.3 Issues & Fixes

Issue	Fix
Potential data leakage from post-delivery columns (Delivery_Duration_Minutes, Delivery_Speed_kmh)	Excluded all post-event variables from the training feature set (X)
Input schema mismatch during web-app execution (KeyError on unseen columns)	Enforced automatic column reordering and smart fallback default-value imputation in predict_delivery_delay()

9.4 Final Review

The team completed a full quality-assurance review covering notebook reproducibility, documentation standards, code cleanups, model evaluation metrics and live cloud-application responsiveness. All requirements were met and signed off for final submission.

Evidence: Final Project Report Document / GitHub Repository. Responsible: Omar Mohamed. Result: a fully integrated, production-ready machine-learning and analytics solution, approved for project submission.

Team & Project Management

10.1 Team Responsibilities

Omar Mohamed Abdelbaqi

TEAM LEADER · ML STRATEGY & MODELING

Project direction and scope; business/research question coordination; ML problem definition and strategy; baseline model and practical implementation; final model review; final integration and quality review; documentation and team coordination.

Ahmed Saad

POWER BI LEAD

Dashboard planning; Power BI data model; DAX measures and KPIs; main dashboard development; final dashboard design.

Bashar Mesbah

PYTHON, DATA CLEANING & EDA

Data understanding, cleaning and preparation; exploratory data analysis; statistical analysis; Python visualizations; EDA findings.

Kyrrillos

POWER BI ANALYSIS & VALIDATION

Power BI data preparation; metrics and analysis; dashboard visuals; Python vs. Power BI validation; dashboard insights.

Yasmin

ML IMPLEMENTATION & STREAMLIT

ML dataset preparation; feature engineering; candidate model training and evaluation; model comparison and final preparation; Streamlit implementation.

10.2 Important Decisions

DECISION — SEPTEMBER 19, 2026

Decision: selected Random Forest Classifier as the final deployment model and excluded post-delivery metrics from the feature space.

Reason: Random Forest achieved the highest overall accuracy (>91%) and ROC-AUC (>95%) while remaining robust against overfitting; excluding duration/speed metrics was necessary to eliminate data leakage. **Impact:** ensured a reliable, realistic model that predicts delivery delays pre-dispatch without reliance on future data. **Decided by:** Omar Mohamed.

10.3 Blockers & Problems

Problem: KeyError thrown during Streamlit inference when passing partial order input dictionaries — the pipeline expected all 15 raw training columns, but the UI form only collected key operational inputs. **Action:** the inference function was updated to automatically impute non-critical missing columns with pre-calculated medians/modes before feature ordering. **Status:** fully resolved (see also Section 07.3 / 09.3).

10.4 Mentor / Instructor Feedback

Feedback: ensure the model target is realistic, free of data leakage, and accompanied by interactive deployment and operational dashboards. **Required action:** construct engineered delay-risk features, validate zero leakage, deploy a Streamlit web app, and design a Power BI operational dashboard. **Action taken:** built the compound Traffic_Delay_Risk feature, strictly removed duration/speed columns, deployed the live app on Streamlit Cloud, and completed the Power BI data model and DAX metrics. **Status:** Completed.

10.5 Final Presentation Structure

- 1 Problem Statement
- 2 Data Sources
- 3 Data Architecture / Data Schema
- 4 Data Cleaning
- 5 Machine Learning and Model Evaluation (Streamlit demo shown here)
- 6 Dashboard and Inferred Insights
- 7 Conclusion

Key Takeaways & Conclusion

Together, the two source documents describe one continuous body of work: a single flat file of simulated food-delivery orders, profiled, cleaned and enriched into `final_clean_delivery_data.csv`, then developed along two parallel, complementary tracks.

11.1 The Machine Learning Track

Delivered a leakage-free, deployment-ready Random Forest classification pipeline predicting `Is_Delayed` with over 90% accuracy and over 95% ROC-AUC, packaged with a Streamlit-ready inference function and deployed as a public web application — resolving a real integration issue (partial-input `KeyErrors`) along the way.

11.2 The Power BI Track

Normalized the same data into a star schema of one fact table and four dimensions, adding external reference data (item cost, city coordinates) that the source file lacked, and surfaced the result through a themed, four-page interactive dashboard. Without the added cost attribute there would be no profit measure; without the city dimension there would be no map; without the generated calendar there would be no reliable time analysis.

11.3 What the Combined Analysis Shows

- A concentrated product mix, with two items driving nearly half of top-line revenue.
- A substantial cancellation rate (9.8%) and an even larger delay rate (43%), both flagged as priority levers.
- An evenly distributed geographic and payment profile, consistent with the dataset's simulated nature.
- A clear, statistically-supported association between distance and delivery duration (confirmed independently by both the Python EDA in Section 04 and the Power BI scatter plot in Section 06.4).
- A leakage-free predictive model that operationalizes these patterns into a pre-dispatch delay warning, ready for integration into the same dashboard's operational view.

READ TOGETHER

The Executive Overview answers the commercial questions — what was sold, at what value, through which channels, at what profitability, with what cancellation rate. The Delivery Performance and Geography pages answer the operational questions — how long deliveries take, how far they travel, how often they're delayed, and how the fleet is used. The predictive model closes the loop by turning those retrospective patterns into a forward-looking, pre-dispatch delay warning. Presented through a consistent, themed interface, the combined project gives a reader a single, well-structured picture of both the results of the delivery business and the conditions that produce them.

11.4 Submission Status

Item	Status
Overall Project Status	Completed
Final Documentation	Completed (this document)
ML Pipeline & Model	Completed — Random Forest, >90% accuracy
Power BI Dashboard	Completed — 4-table star schema, multi-page report
Streamlit Application	Completed — live public deployment
Next Major Task	Final project presentation & review

Final Submission Status

Ready for submission

Glossary

Data Leakage

Accidentally training a model on information that would not be available at prediction time — here, post-delivery fields like actual duration or speed. Eliminated by excluding those columns from the feature set (Section 05.3, 09.3).

Star Schema

A dimensional model with one central fact table (measurable events) surrounded by dimension tables (descriptive attributes), connected by one-to-many relationships (Section 03.6).

Fact Table

The central table holding keys and measurable values for each event — here, FactOrders holding order-level metrics.

Dimension Table

A table of descriptive attributes referenced by the fact table via a surrogate key — e.g. DimItems, DimCity, DimDate, DimDrivers.

Surrogate Key

A generated identifier (e.g. an index column) used as a stable primary key, independent of the underlying text value (e.g. Item_id instead of Item_Name).

ColumnTransformer

A scikit-learn construct that applies different preprocessing (scaling, encoding) to different columns within a single pipeline.

ROC-AUC

Area under the Receiver Operating Characteristic curve — a threshold-independent measure of a binary classifier's ability to separate the two classes.

One-Way ANOVA

A statistical test comparing the means of a numeric variable (e.g. delivery duration) across more than two categorical groups (e.g. traffic levels).

Pearson Correlation

A statistic measuring the strength and direction of a linear relationship between two numeric variables.

DAX

Data Analysis Expressions — the formula language used for Power BI measures; the specific DAX code behind this project's KPIs was not supplied with the source materials (Section 03.7, 06.1).

Power Query (M)

Power BI's data-transformation engine and language, used here for all cleaning, joining and normalization steps (Section 03).

Appendix — Reference Material

13.1 Selected Power Query (M) Code

Selected steps from the supplied Power Query scripts, evidencing the transformations documented in Section 03.

FactOrders — derived business columns

```
#"Added Custom" = Table.AddColumn(#"Removed Columns", "Total Sales",
    each [Quantity]*[items.Price]),
#"Added Custom1" = Table.AddColumn(#"Expanded DimItems1", "Profit",
    each [Total Sales]-([DimItems.cost]*[Quantity])),
```

FactOrders — normalization joins and column reduction

```
#"Merged Queries" = Table.NestedJoin(#"Changed column type", {"Item_Name"},
    DimItems, {"Item_Name"}, "DimItems", JoinKind.LeftOuter),
#"Merged Queries1" = Table.NestedJoin(#"Reordered Columns2", {"City"},
    Dimcity, {"City"}, "Dimcity", JoinKind.LeftOuter),
#"Removed Columns1" = Table.RemoveColumns(#"Reordered Columns3",
    {"Item_Name", "items.Price", "City", "Driver_Vehicle",
    "Is_Weekend", "Driver_Order_Count"}),
```

DimItems — deduplication, surrogate key and cost

```
#"Removed Other Columns" = Table.SelectColumns(#"Changed column type",
    {"Item_Name", "Total_Price"}),
#"Removed Duplicates" = Table.Distinct(#"Removed Other Columns", {"Item_Name"}),
#"Added Index" = Table.AddIndexColumn(#"Replaced Value1", "Index", 1, 1, Int64.Type),
#"Renamed Columns" = Table.RenameColumns(#"Reordered Columns",
    {{"Index", "Item_id"}, {"Total_Price", "Price"}}),
#"Duplicated Column" = Table.DuplicateColumn(#"Renamed Columns", "Price", "Price - Copy"),
#"Renamed Columns1" = Table.RenameColumns(#"Duplicated Column",
    {{"Price - Copy", "cost"}}),
```

DimDate — dynamic range and Egyptian weekend flag

```
StartDate = List.Min(FactOrders[Order_Date]),
EndDate = List.Max(FactOrders[Order_Date]),
DateList = List.Dates(StartDate,
    Duration.Days(EndDate - StartDate) + 1, #duration(1, 0, 0, 0)),
AddIsWeekend = Table.AddColumn(AddDayOfWeek, "Is_Weekend",
    each if [DayOfWeek] = "Friday" or [DayOfWeek] = "Saturday"
    then true else false, type logical)
```

Dimcity — typed external reference table

```
#"Changed column type" = Table.TransformColumnTypes(Source,
    {{"City_id", Int64.Type}, {"City", type text},
    {"Latitude", type number}, {"Longitude", type number}})
```

13.2 Evidence & File Index

Track	File / Notebook
Data Understanding	01_Data_Understanding.ipynb, Data Dictionary

Data Cleaning	Cleaning notebook, final_clean_delivery_data.csv
EDA & Statistics	03_EDA_Statistical_Analysis.ipynb
Predictive Modeling	Delivery_Delay_ML_Notebook.ipynb, final_delivery_delay_model.pkl
Streamlit App	app.py
Power BI Dashboard	Order_Delivery_Analytics.pbix

13.3 Submission Checklist (as logged)

- Final documentation completed
- Final dataset prepared
- Analysis code / notebooks completed
- Power BI dashboard completed
- Machine learning work completed
- Streamlit application completed
- GitHub repository updated
- Final presentation completed
- Final team review completed

Final project links (GitHub repository, Power BI file, Streamlit application, final dataset) were left as placeholders ("Add link here") in the team's progress log, aside from the Streamlit app and GitHub links captured in Section 07.4.